

JUnit Testing Exercises

Exercise 1: Setting Up JUnit

Scenario:

You need to set up JUnit in your Java project to start writing unit tests.

Steps:

1. Create a new Java project in your IDE (e.g., IntelliJ IDEA, Eclipse).
2. Add JUnit dependency to your project. If you are using Maven, add the following to your pom.xml:

```
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>4.13.2</version>
<scope>test</scope>
</dependency>
```

3. Create a new test class in your project.

Code:

```
<dependencies>
  <dependency>
    <groupId>junit</groupId>
    <artifactId>junit</artifactId>
    <version>4.13.2</version>
    <scope>test</scope>
  </dependency>
</dependencies>
```

Exercise 2: Writing Basic JUnit Tests

Scenario:

You need to write basic JUnit tests for a simple Java class.

Steps:

1. Create a new Java class with some methods to test.
2. Write JUnit tests for these methods.

Code:

```
public class Calculator {
    public int add(int a, int b) {
        return a + b;
    }
    public int subtract(int a, int b) {
        return a - b;
    }
    public int multiply(int a, int b) {
        return a * b;
    }
    public int divide(int a, int b) {
        return a / b;
    }
}
```

```
}  
}
```

Exercise 3: Assertions in JUnit

Scenario:

You need to use different assertions in JUnit to validate your test results.

Steps: 1. Write tests using various JUnit assertions.

Solution Code:

```
public class AssertionsTest {  
    @Test  
    public void testAssertions() {  
        // Assert equals  
        assertEquals(5, 2 + 3);  
        // Assert true  
        assertTrue(5 > 3);  
        // Assert false  
        assertFalse(5 < 3);  
        // Assert null  
        assertNull(null);  
        // Assert not null  
        assertNotNull(new Object());  
    }  
}
```

Here it will show output has 1/1 testcase passed

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and

Teardown Methods in JUnit

Scenario:

You need to organize your tests using the Arrange-Act-Assert (AAA) pattern and use setup and teardown methods.

Steps:

1. Write tests using the AAA pattern.
2. Use @Before and @After annotations for setup and teardown methods

Code:

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
}
```

